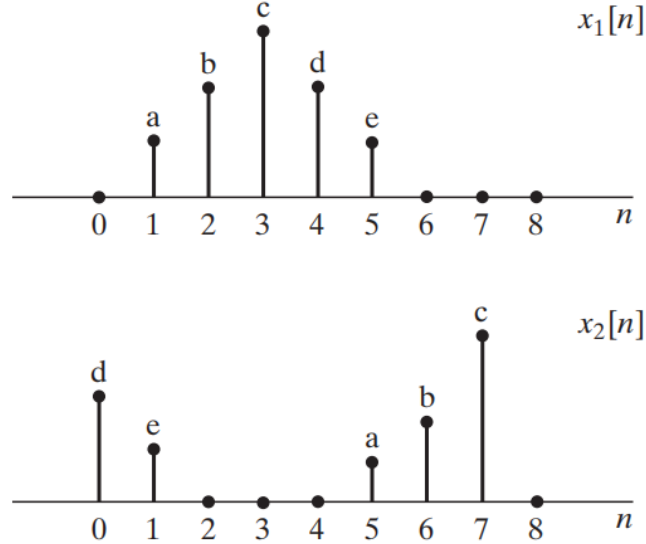
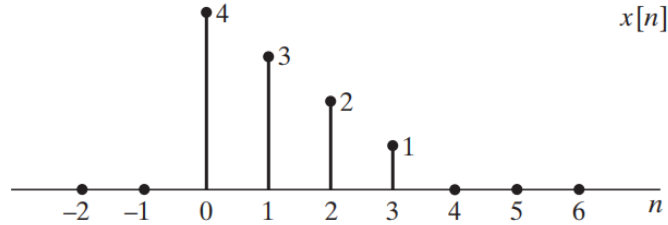


Problem Set 3

1. The two eight-point sequences $x_1[n]$ and $x_2[n]$ shown below have DFTs $X_1[k]$ and $X_2[k]$, respectively. Determine the relationship between $X_1[k]$ and $X_2[k]$.



2. Consider the real finite-length sequence $x[n]$ shown below.



- (a) Sketch the finite-length sequence $y[n]$ whose six-point DFT is

$$Y[k] = W_6^{5k} X[k],$$

where $X[k]$ is the six-point DFT of $x[n]$.

- (b) Sketch the finite-length sequence $w[n]$ whose six-point DFT is

$$W[k] = \text{Im}\{X[k]\}.$$

- (c) Sketch the finite-length sequence $q[n]$ whose three-point DFT is

$$Q[k] = X[2k + 1], \quad k = 0, 1, 2.$$

3. Consider two signals $p[n] = \{1, 2, 0, 1\}$ and $q[n] = \{2, 2, 1, 1\}$.

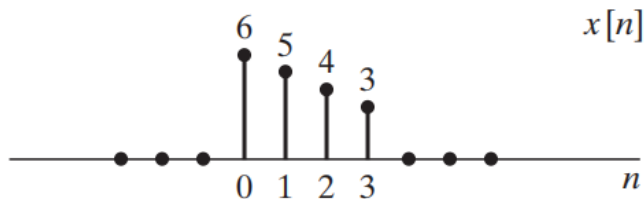
- (a) Calculate the DFT of $p[n]$ and $q[n]$.
 (b) Calculate the circular convolution between $p[n]$ and $q[n]$.

4. The finite-length sequence $x[n]$ is shown below. Sketch the sequences

$$x_1[n] = x[((n-2))_4], \quad 0 \leq n \leq 3,$$

and

$$x_2[n] = x[((-n))_4], \quad 0 \leq n \leq 3.$$



5. A general 1024-point DFT is to be computed. What is the total number of required complex multiplications:

- (a) if the computations are done using the DFT directly, and
- (b) if the computations are done using an FFT?

Results can be written in the form of integer powers of 2.

6. Find the DFT of the signal $x[n] = \{1, 2, 3, 4, 4, 3, 2, 1\}$.

- (a) using decimation-in-time FFT algorithms. Show the flow graph.
- (b) using decimation-in-frequency FFT algorithms. Show the flow graph.